

Aarul Dhawan

Email | LinkedIn | GitHub

EDUCATION

University of Illinois Urbana-Champaign

Masters in Computer Science | Networks and Distributed Systems Concentration May 2025 - May 2026
Bachelors in Computer Science and Statistics (BS) | GPA: 3.93 | Highest Distinction Aug 2022 - May 2025
Relevant Coursework: Deep Learning for Computer Vision, System Programming, Computer Architecture, Database Systems, Data Structures, Programming Languages and Compilers, Statistical Modeling, Algorithms

EXPERIENCE

Amazon Web Services

May 2025 - Aug 2025

Software Development Engineer Intern — S3 Express

Seattle, WA

- Re-architected a legacy Java load generator into a high-performance Rust system for S3 Express OneZone, improving throughput by **over 400%** and achieving consistent **sub-10ms p99 latency**.
- Added first-time support for benchmarking across multiple SDKs, enabling cross-language performance validation of S3 Express buckets in real-world customer scenarios.
- Deployed on AWS EKS with autoscaling to dynamically scale **hundreds of pods in under 60s**, while publishing **p50–p99.9 latency, error rates, and TPS metrics** into CloudWatch dashboards.
- Partnered with a customer team to onboard onto Rust SDK logging, reducing client-side latency overhead and enabling deeper observability of S3 operations in production.

Epivara

Sept 2024 - Jan 2025

Software Engineer Intern

UIUC Research Park, IL

- Created an NLP-powered graph generation tool that interprets natural language queries to generate visualizations from experimental data stored in a SQL database, streamlining data exploration for users..
- Led the establishment of the company's software department, establishing core systems, tooling, and engineering practices to support AI-driven research workflows

Epic Systems

May 2024 - Aug 2024

Software Developer Intern

Madison, WI

- Developed a cross-platform AI assistant using C#, Python, and React to compare echocardiogram studies, improving diagnostic efficiency in clinical settings.
- Estimated to save **\$3.4M+ annually** in labor costs for cardiologists across the U.S. healthcare system.
- Designed and implemented an LLM Evaluation Metric for patient summarization initiatives, and presented results to 200+ attendees while training 30+ Application Leads on adoption.

Disruption Lab, (Client: AMD)

Aug 2023 - May 2024

Software Engineer

Champaign, IL

- Collaborated with sound engineers to leverage ML models for audio enhancement of AMD Audio Products.
- Modified data pipeline using dynamic mixing and noise augmentation techniques, increasing model robustness by 2.1x in noisy environments.
- Trained and fine-tuned ML models on AWS SageMaker across 200+ hours of audio data, reducing training time by 18% via pipeline optimizations.

TechSur Solutions

Jun 2023 - Aug 2023

Software Developer Intern

Herndon, VA

- Built a document recommendation model using TF-IDF vectorization with Scikit-learn and Pinecone hosted on Google Cloud Platform (GCP) to surface relevant document content for users.
- Practiced **Agile development methodology**: contributed in daily standups, collaborated cross-functionally, and consistently met sprint deliverables.

PROJECTS

Arcade Karaoke | Brown Hackathon 2025 | AWS Award Winner

Python, React, WebSockets

Multiplayer karaoke game with real-time scoring based on pitch analysis using Librosa.

Learning Management Assignment Collector (LeMAC)

Python, React, Selenium

Aggregates assignments from Canvas and PrairieLearn into a unified view.

ADDITIONAL INFORMATION

Languages/Frameworks: Java, Rust, Python, C++, C#, C, JavaScript, React, PostgreSQL, R, MySQL

Honors: Penn State College of Engineering Dean's List (2/2 Semesters), UIUC Dean's List (1/1 Semesters)